# Review

Last time, we solved differential equations of the form $\frac{dy}{dt} = f(t)$. We solve these by _____ $f(t)$.

**Question** Solve the initial value problem $\frac{dy}{dx} = 3t^2$ with $y(1) = 0$.

# More Complex Differential Equations

Consider the differential equation $y'(t) = y(t)$. Complete the blank: The solution to this differential equation is a function $y(t)$ whose derivative is equal to _____.

What function $y(t)$ satisfies the sentence you just wrote down? Plug in to both sides of the equation to check it.

Is the function $y(t) = 2 + e^t$ also a solution?

Can you figure out another function that solves the equation? Check it!

Can you write down the general solution to this equation?

Consider the differential equation $y'(t) = 2y(t)$. Complete the blank: The solution to this differential equation is a function $y(t)$ whose derivative is equal to _____.

Can you find one solution to this equation?

Can you find another solution by adding a constant to your solution above?

Write down the general solution to this equation.

Solve the initial value problem $y'(t) = 2y$, $y(0) = 3$.

The next three questions below are about the differential equation $\frac{dy}{dx} = x + y$.

1. Why is the solution to this *not* $y(x) = \int x + y \, dx$? In fact, why does that integral not make any sense?

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2. (a) Check if $y(x) = \frac{x^2}{2}$ is a solution to this equation.

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(b) Check if $y(x) = e^x - x - 1$ is a solution.

(c) Check if $y(x) = e^x - x + 1$ is a solution.

(d) Check if $y(x) = 2e^x - x - 1$ is a solution.

3. ☐Regrade (a) Is $y(x) = e^x - x + C$ a solution for all values of $C$? See the previous question for an easy answer to this!

(b) What about $y(x) = Ce^x - x - 1$? Is that a solution for all values of $C$? The previous question shows it is for $C = 1$ and $C = 2$. Use the same idea as you used in Q2 above to check if this is indeed the general solution of $\frac{dy}{dx} = x + y$.

(c) Solve the initial value problem $\frac{dy}{dx} = x + y$, $y(0) = 4$. Use your general solution from the previous part.

## Higher Order Differential Equations

The next two questions are about the differential equation $\frac{d^2y}{dx^2} = -9y$.

4. (a) What is the order of this equation?

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(b) Check that both $\sin(3x)$ and $\cos(3x)$ are solutions to this equation.

(c) Check that both $2\sin(3x)$ and $3\cos(3x)$ are solutions to this equation.

(d) Check that $2\sin(3x) + 3\cos(3x)$ is a solution to this equation.

5. (a) Can you write down the general solution of this equation?

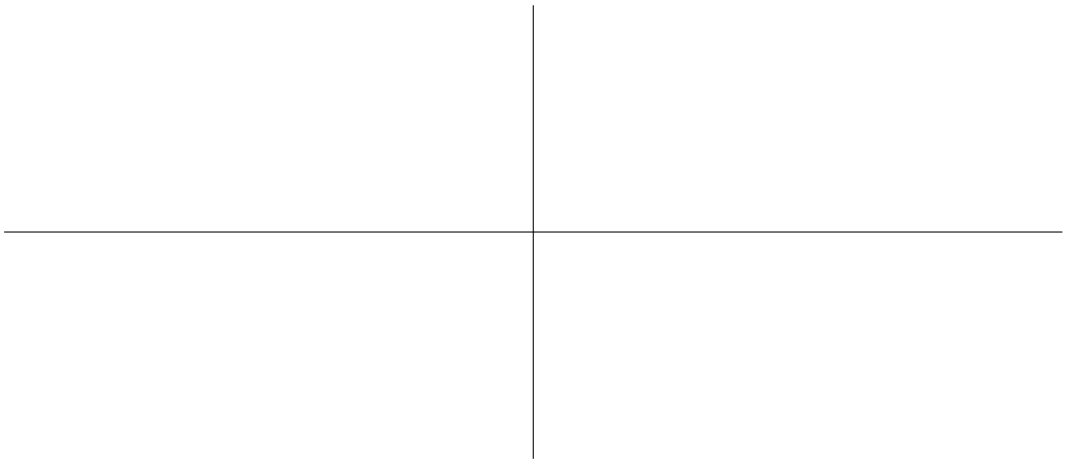
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(b) Solve the initial value problems:

i. $\frac{d^2y}{dx^2} = -9y$, $y(0) = 1$, $y\left(\frac{\pi}{6}\right) = 0$.

ii. $\frac{d^2y}{dx^2} = -9y$, $y(0) = 1$, $y'(0) = 2$.

(c) Graph the two solutions from the previous part below.



In general, an $n^{\text{th}}$ order differential equation has a general solution with $n$ aribitrary constants in it.

As you can see, the solutions to a differential equation may not related to each other as simply as the ones we saw at the beginning of this worksheet!

# What we can and can't do (for now)

- The only differential equations we can currently solve analytically–that is, without (much) guessing–are those of the form

$$\frac{dy}{dt} = f(t), \text{ or} \frac{d^2y}{dt^t} = f(t).$$

  We solve these by _____ $f(t)$ once or twice.

- We cannot (for now) solve equations like

$$\frac{dy}{dt} = y^2.$$

  We will do (some of) these later in the semester. For now, given a solution, we can check it.

- We cannot solve second (or higher) order differential equations other than $\frac{d^2y}{dt^2} = f(t)$. We will leave other second order equations for a later course.

- We can also check whether a given function solves a particular differential equation.

- We can use a general solution to solve initial value problems, given some initial conditions.

# More Questions

6. By considering the DEs on the first two pages of this worksheet, find the general solution of the differential equation $\frac{dy}{dx} = ky$, where $k$ is a fixed constant (Hint: up there, we considered $k = 1$ and $k = 2$). □Regrade

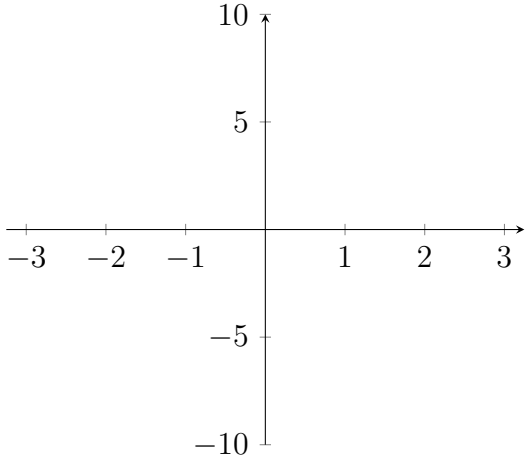
(a) Use your solution to solve the following initial value problems:

i. $\frac{dy}{dt} = ky$, $y(0) = 2$.

ii. $\frac{dy}{dt} = ky$, $y(\ln 2) = 2$.

iii. $\frac{dy}{dt} = ky$, $y(1) = 2$.

(b) Graph solutions to the equation $\frac{dy}{dx} = 3y$ with $y(0) = 1$, $y(0) = 2$, and $y(0) = -1$ on the same set of axes.



7. Is the function $y = -\frac{1}{x}$ a solution to the equation $\frac{dy}{dx} = y^2$? Check that $y = \frac{1}{x}$ is *not* a □Regrade solution. What about $y = -\frac{1}{x+c}$? Find a solution with $y(1) = -\frac{1}{2}$.

8. Find a function of the form $y = x^n$ that is a solution to the differential equation □Regrade $\frac{1}{2}x\frac{dy}{dx} = y$. (Hint: Assume $y = x^n$ and use the method we used above to check solutions to DEs.)